## Engine Intake Deflector System – General Data

1. General

Refer to Figure 1 for details.

The engine intake deflector system prevents the entry of snow and rain into the engine compressor inlet duct by diverting the particles aft over the oil cooler honeycomb and out through the inertial separator exit duct. Each engine intake is equipped with an identical system which consists of a deflector, two actuators, an air valve, two spring-loaded lock levers, a release lever, microswitches, an exit duct door and an intake deflector switch (common to both systems). The system is controlled by the air intake deflector switch, in the flight compartment, which, when switched to EXTEND, energizes the air valve and allows engine bleed air pressure into the two actuators. The actuators extend fully, lower the attached deflector, and release the tension on the exit duct door cables allowing the spring-loaded exit duct door to open. Spring-loaded lock levers hold the deflector in the extended position. When the INTAKE DEFLECTOR switch is released the air valve is deenergized and relieves the actuators to atmosphere. On retraction, (RETRACT selection) the air valve is energized and the actuators extend relieving the pressure on the lock levers. The release lever solenoid is also energized, rotates a cam to release the locking levers, and operates a microswitch which deenergizes the air valve. The deenergized air valve shuts off bleed air and vents the air from the actuators through the valve, allowing the actuator return spring to retract the deflector. The control circuit is powered from the 28 Volt DC supply and is protected by a 10–ampere INTAKE DEFLECTORS circuit breaker on the main circuit breaker panel. The deflector cannot be selected to an intermediate position. Indication that the intake deflectors have extended and locked in place is provided by display of the text 'A/I' (for anti-ice) on the outboard portion of each engine parameter display. It is operated by two microswitches connected in series, one at the intake deflector and one at the exit duct door. An air line filter is installed in each AN fitting connection at the air valve.

A. Engine Intake Deflector Actuators

The engine intake deflector actuators are screwed into adapters mounted in the inlet duct and are connected by turnbuckle type piston rods to the outer edges of the deflector. Each actuator contains a piston and return spring.

B. Engine Intake Deflector

The engine intake deflector is a louvered plate hinged, at the forward end, to the roof of the intake. The plate is connected to two actuators at the aft end, and mechanically linked by cable to the exit duct door. A striker is attached at the hinge to actuate a microswitch in the indicator circuit.

C. Engine Intake Deflector Air Valve

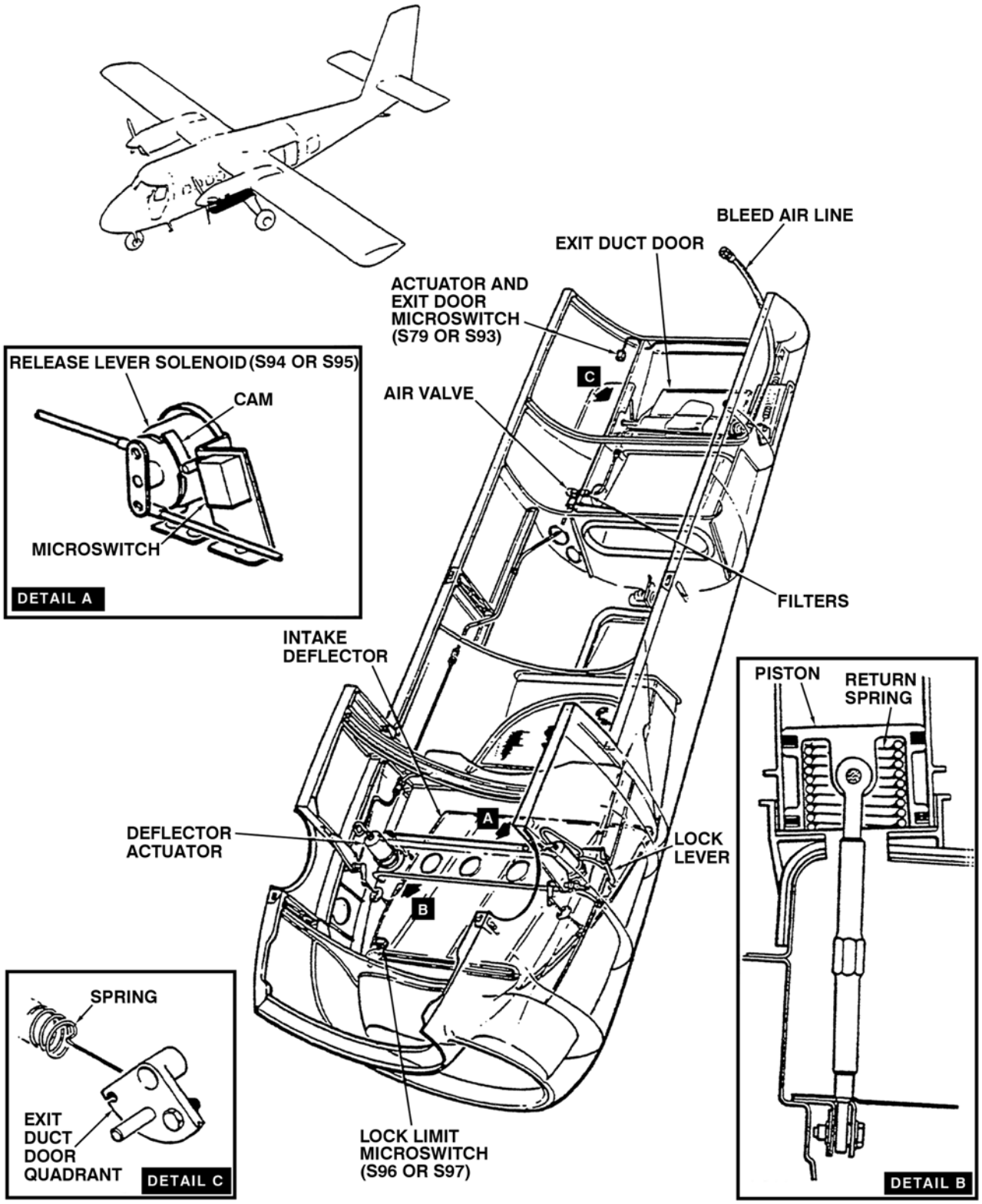
The engine intake deflector air valve is mounted in the lower cowling to the rear of the actuators. The valve is electrically-operated and controls the flow of air pressure in the air line taken from the bleed air system line of the engine.

D. Engine Intake Deflector Locking Levers

The engine intake deflector locking levers are mounted, one on each side of the inlet duct, and protrude through slots into the duct. They are spring-loaded and connected by rods to the release lever which rotates and operates a microswitch to deenergize the air valve, which vents to atmosphere.

E. Engine Intake Deflector Solenoid-Operated Release Lever

The engine intake deflector release lever is mounted above the inlet duct between the locking levers and is connected to them by adjustable rods. The release lever is operated by a solenoid which is controlled by the intake deflector switch.




Engine Intake Deflector System
Figure 1

F. Engine Intake Deflector Switch

The INTAKE DEFLECTOR switch is located on the co-pilot's switch panel and controls the deflectors of both engines. The switch is labeled INTAKE DEFLECTOR and has positions marked RETRACT and EXTEND. It is spring-loaded to the centre “off” position.

G. Engine Intake Deflector Indicators

Indication that the intake deflectors have extended and locked in place is provided by display of the text 'A/I' (for anti-ice) on the outboard portion of each engine parameter display.

2. Troubleshooting

A. Extend Cycle

| TROUBLE | POSSIBLE CAUSE | CORRECTION |
|---|---|---|
| Intake deflector fails to extend. | Release lever solenoid microswitch not correctly adjusted on its operating cam. | Adjust position of microswitch. |
| | Pistons jamming in operating cylinders. | Remove pistons and inspect cylinders for signs of scoring and dirt. Ensure each groove in each piston has two piston rings and a corrugated expander; that rings are loose enough to rotate by hand, and that gaps in rings are approximately 180° apart from each other. Lightly lubricate piston rings with silicone grease at assembly. |
| | Dirt in air valve. | Replace air valve or clean air valve filter. |
| | Electrical failure. | Loss of voltage on right bus (could be open bus tie switch). Faulty bus tie switch relay. Faulty circuit breaker. Check bonding between lower cowl and airframe. |
| Snow separator exit duct door not fully open. (With intake deflector in EXTEND position). | Tension springs not hooked up. | Attach tension springs to door operating boss. |
| | Door control cables adjusted too short. | Adjust cables. |
| Engine intake deflector position is indicated by A/I (Anti-Ice) displayed in the engine parameter window on the pilot's and co-pilot's PFD. | Deflector position and exit door microswitches require adjustment. | Position both deflector and exit door microswitches so that they are actuated with intake deflector in EXTEND position. |
| | Open circuit to indicator. | Rectify wiring as necessary. |
| | Faulty indicator. | Replace indicator. |

B. Retract Cycle

| TROUBLE | POSSIBLE CAUSE | CORRECTION |
|---|---|---|
| Intake deflector fails to retract. | Screws securing cam to release lever solenoid protruding through solenoid mounting plate. | Replace any screws restricting solenoid rotational travel with shorter ones. |
| | Lock levers incorrectly adjusted or sticking. | Ensure locking levers are correctly adjusted.To adjust the locking lever position, vary the amount of exposed thread on the threaded rod end. Check that locking levers move freely.<br>Note Ensure adequate threads are visible in inspection hole. |
| | Defective release lever solenoid. | Replace solenoid if it does not rotate. |
| | | **CAUTION DO NOT ACTUATE RETRACT SWITCH FOR MORE THAN 5 SECONDS AT ANY ONE TIME.** |
| | Dirt in air valve. | Replace air valve or clean air valve filter. |
| | Broken piston spring(s). | Remove pistons and replace broken piston springs. |
| | Pistons jamming in operating cylinders. | Remove pistons and replace broken piston springs. |
| | Electrical failure. | Loss of voltage on right bus (could be open bus tie switch). Faulty bus tie switch relay. Faulty circuit breaker. |
| Exit duct door not fully closed (with intake deflector in RETRACT position). | Too much play in control cables. | Adjust cables. |
| Intake deflector position indicator is displayed with retract cycle completed. | Deflector position and exit duct door position microswitches not correctly adjusted. | Adjust microswitches. |
| | Faulty microswitch(es). | Replace microswitches, as necessary. |
| | Indicator circuit shorted. | Rectify wiring as necessary. |
| | Faulty indicator. | Replace indicator. |

3. Adjustment/Test

Equipment required:

- Pressurized air or nitrogen bottle
- Calibrated Pressure Regulator
- Shut-off valve
- Flexible tubing
- MS24392D4 Union (or equivalent) and
- External Electrical Power Supply

A. Adjustment

Refer to Figure 1 for details.

1) Remove Exit Duct Door Microswitch (S79 or S93) and Actuator from the lower cowling.

2) Bend up end of Actutator Arm to 45 degrees at 0.25” to clear the forward end of the operating arm.

3) Reinstall the Exit Duct Door Microswitch (S79 or S93) and Actuator on the lower cowling.

4) Ensure that Exit Duct Door Actutator Arm clears operating lever.

5) Disconnect supply line downstream of the nacelle air valve, remove in-line air filter, and attach shop air using MS24392D4 union and regulator.

6) Set regulated air pressure to 35±5 psi and ensure intake deflector moves smoothly to an extend position.

7) Reduce air pressure until intake deflector is locked in fully extended position.

8) Disconnect shop air.

9) Set Lock Limit Microswitch (S96 or S97) to actuate with intake deflector in the fully extended and locked position.

**CAUTION** **MAKE SURE THAT MICROSWITCH ACTUATOR TAB HAS SUFFICIENT TRAVEL IN THIS POSITION TO PREVENT DAMAGE.**

10) Reconnect the shop air using MS24392D4 union and regulator. Slowly increase the air pressure until the lock levers are unloaded (air screen fully extended).

11) Verify exit duct door is fully open.

**CAUTION** **MAKE SURE EXIT DUCT DOOR DOES NOT GET STUCK INSIDE DUCT WHEN FULLY OPEN.**

12) Manually set the Release Lever Solenoid Limit Microswitch (S94 or S95) to actuate with intake deflector in fully extended position and levers in the retracted position..

13) With Lock Limit Microswitch (S96 or S97) Exit Duct Door Microswitch (S79 or S93) in correct rigging position, check AI indication in PFD.

**CAUTION** **MAKE SURE THAT MICROSWITCH ACTUATOR TAB HAS SUFFICIENT TRAVEL IN THIS POSITION TO PREVENT DAMAGE.**

14) Observe that door is clear of locking levers then manually rotate control solenoid to stow levers in unlocked position.

15) Hold the Release Solenoid Limit switch (S94 or S95) open, reduce the shop air. Check that door fully retracts.

16) Disconnect shop air, remove MS24392D4 union, reinstall in-line air filter and the supply line at the Bleed Air Solenoid Valve.

17) Carry out leak check of installed tube fittings and door actuators.

18) Do a function test as per Para 3.B.

B. Function Test

Equipment required:

- Pressurized air or nitrogen bottle
- Calibrated Pressure Regulator
- Shut-off valve
- Flexible tubing

1) Extend Cycle

a) Lower bottom engine cowl.

b) Disconnect intake deflector air pressure line at firewall bulkhead, cap firewall to engine line.

c) Route flexible line from pressure bottle through access hole in top cowl and connect to intake deflector air pressure line.

d) Raise lower engine cowl and secure latches.

e) With regulator closed, open pressurized bottle shut-off valve and set regulator to 35 ± 5 psi.

f) Connect external electrical supply to aircraft.

g) Ensure INTAKE DEFLECTOR circuit breaker is engaged.

h) Hold INTAKE DEFLECTOR switch in EXTEND position until intake deflector displays 'A/I' (Anti-Ice) on the PFD.

i) Ensure intake deflector moves smoothly to extend position and locks. Time 3 seconds approximately.

j) Check exit duct door is fully open.

2) Retract Cycle

**CAUTION** **DO NOT ACTUATE RETRACT SWITCH FOR MORE THAN 5 SECONDS AT ANY ONE TIME.**

a) Hold INTAKE DEFLECTOR switch in RETRACT position until intake deflector displays 'A/I' (anti-ice).

b) Ensure intake deflector becomes unlocked and retracts flush with top of intake duct.

c) Check exit duct door is fully closed.

d) Switch off and disconnect electrical power.

e) Close off pressurized bottle shut-off valve and disconnect from intake deflector line.

f) Remove blanking cap and connect intake deflector air pressure line at firewall bulkhead.

C. Secure Intake Door Locking Lever Rod

1) Do function test, as per Para 3.B.

2) After function test, to prevent rotation of locking lever actuating rod apply Grade M Loctite to threads at the end of intake door locking lever actuating rods.

3) Repeat function test, as per Para 3.B, to confirm proper operation.

**This Page Intentionally Left Blank**